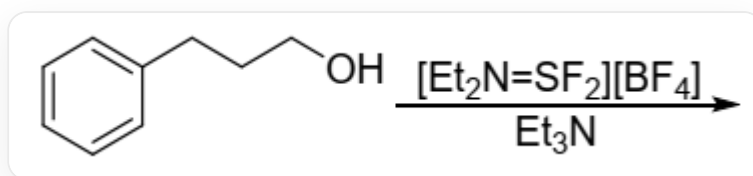


Question

For the following organic reaction, select the incorrect description regarding its organic product.



The image shows a rightward horizontal arrow. On the left side of the arrow is an organic molecule with the structure c1ccccc1CCCO. Above the arrow is the text $[\text{Et}_2\text{N}=\text{SF}_2][\text{BF}_4]$, and below the arrow is the text Et_3N .

- A.** Under this reaction condition, the mass fraction of the halogen in the generated halogenated hydrocarbon molecule is 13.75%.
- B.** Under this reaction condition, the two atoms with the longest shortest path in the generated ether pass through 16 chemical bonds.
- C.** Under this reaction condition, an organic molecule with a 4-coordinate sulfur atom was generated, and the total number of atoms in the molecule is 45.
- D.** Under this reaction condition, an organic compound without a benzene ring was generated, which contains atoms of six different elements.
- E.** Under these reaction conditions, among the benzene-ring-containing products generated, the product with the highest relative molecular mass exceeds 300.
- F.** Under these reaction conditions, among the generated products, there are organic compounds devoid of benzene rings, with relative molecular masses ranging between 130 and 140.

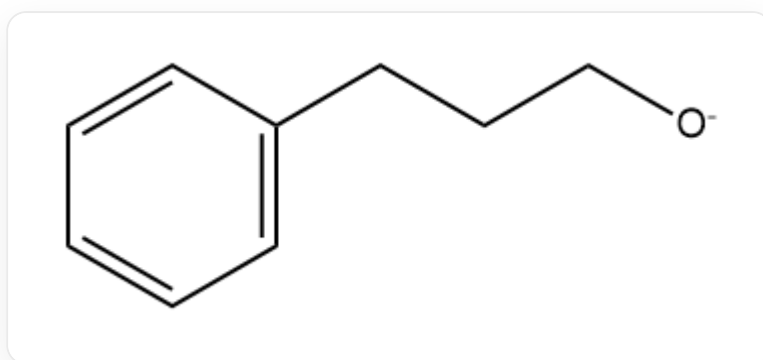
- G.** Under these reaction conditions, there are no two organic products with a relative molecular mass difference of less than 10 between them.
- H.** If the reaction is carried out in acetonitrile solvent, a byproduct containing nitrogen, oxygen, and benzene rings but no fluorine or sulfur will be obtained.

Answer

Correct Answer: **G**

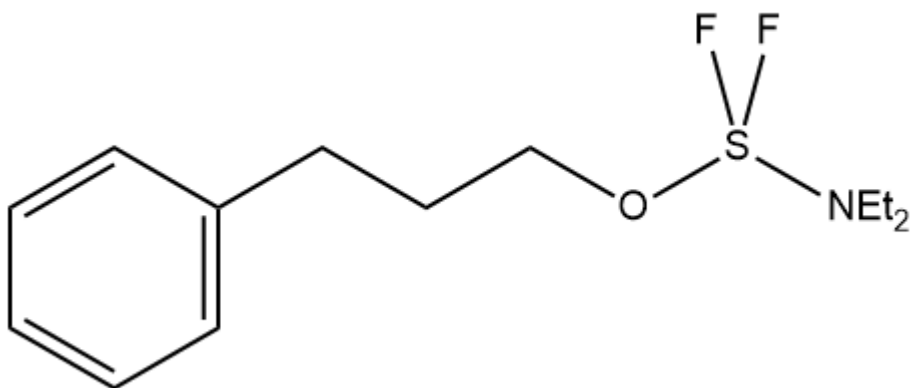
Detailed Explanation

In this reaction system, triethylamine acts as a base to deprotonate the hydroxyl group of the alcohol, yielding an alkoxide intermediate:



c1ccccc1CCC[O-]

In the cation of $[\text{Et}_2\text{N} = \text{SF}_2] [\text{BF}_4]$, the sulfur atom is electrophilic. Upon attack by the alkoxide of the substrate, a neutral intermediate containing tetracoordinated sulfur—with both nitrogen and oxygen ligands—is formed:



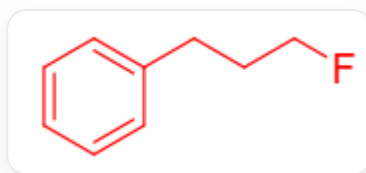
CCN(CC)S(F)(F)OCCCC1CCCCC1

CHECKPOINT

1 PTS

Intermediate CCN(CC)S(F)(F)OCCCC1CCCCC1 generated

This intermediate subsequently undergoes an intramolecular substitution reaction. The possible mechanism involves the lone pair on O attacking the antibonding orbital of the S – F bond, cleaving the S – F bond to form a tight ion pair. The fluoride ion then attacks the carbon bonded to oxygen, replacing the oxygen of the alkoxide to yield a monofluorinated product:



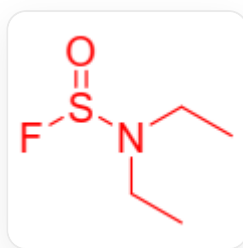
c1ccccc1CCCF

CHECKPOINT

1 PTS

Monofluorinated product c1ccccc1CCCF

Simultaneously, a product containing a sulfur-oxygen bond is formed:



CCN(CC)S(=O)F

CHECKPOINT

1 PTS

Byproduct CCN(CC)S(=O)F departs

These two are the main products under these conditions. The relative molecular mass of the monofluorinated product is $19.00 + 12.01 \times 9 + 1.008 \times 11 = 138.178$, and the mass fraction of fluorine is $\frac{19.00}{138.178} \times 100\% = 13.75\%$. Thus, option **A** is correct.

CHECKPOINT

1 PTS

Relative molecular mass of fluorinated product is 138.178

CHECKPOINT

1 PTS

Mass fraction of fluorine is 13.75%

The other product does not contain a benzene ring, and its relative molecular mass is 139.2, falling between 130 – 140. Thus, option **F** is correct. This molecule contains six elements: C, H, O, N, F, S, so option **D** is also correct.

CHECKPOINT

1 PTS

Product lacks a benzene ring; relative molecular mass is 139.2

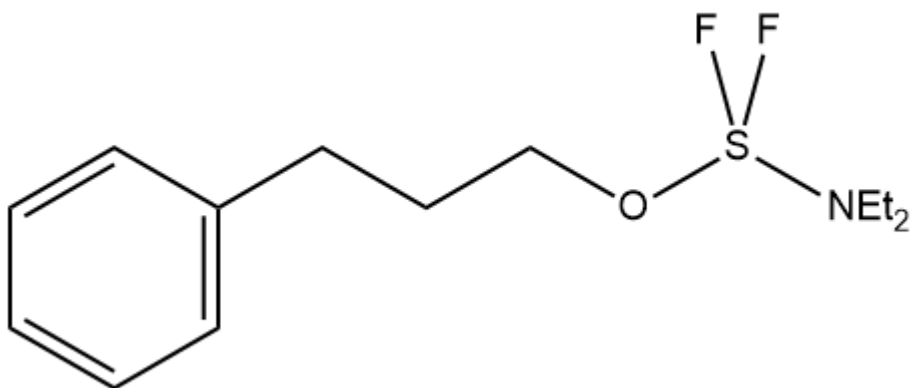
CHECKPOINT

1 PTS

Product contains C, H, O, N, F, S six elements

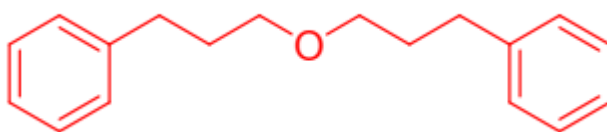
Note that the difference in relative molecular mass between the monofluorinated product and the departing oxygen-containing molecule is $1.022 < 10$, making option **G** incorrect and thus a candidate for exclusion.

The intermediate



CCN(CC)S(F)(F)OCCc1ccccc1

can also be nucleophilically attacked by the alkoxide. If the alkoxide attacks the carbon bonded to oxygen, a stabilized anion leaves by expelling F^- , yielding an ether:



c1ccccc1CCCOCCc2ccccc2

CHECKPOINT

1 PTS

Further nucleophilic attack yields an ether

Its relative molecular mass is 254.356.

CHECKPOINT

1 PTS

Relative molecular mass of ether is 254.356

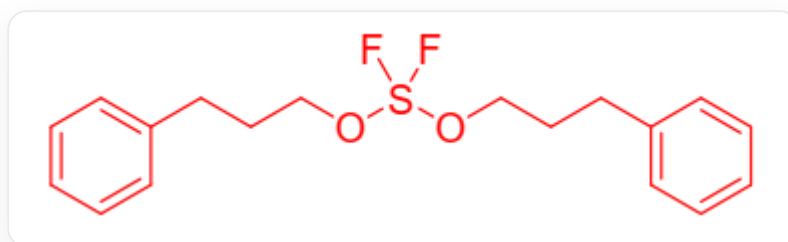
In this molecule, the longest shortest path along chemical bonds occurs between a hydrogen at the *para*-position of one benzene ring and a hydrogen at the *para*-position of the other benzene ring: $\text{H} - \text{C} - \text{C} - \text{C} - \text{C} - \text{C} - \text{C} - \text{C} - \text{O} - \text{C} - \text{C} - \text{C} - \text{C} - \text{C} - \text{C} - \text{C} - \text{H}$, totaling 17 atoms and 16 bonds. Thus, option **B** is correct.

CHECKPOINT

1 PTS

Path spans from *p*-hydrogen of one benzene ring to *p*-hydrogen of the other, with 16 bonds

If the alkoxide attacks the sulfur atom of the sulfur-containing intermediate, a pentacoordinated intermediate forms, expelling a diethylamino anion to yield a byproduct with two oxygens bonded to sulfur:



c1ccccc1CCCOs(F)(F)OCCc2ccccc2

Its relative molecular mass is 340.426, so option **E** is correct.

CHECKPOINT

1 PTS

Product relative molecular mass is 340.426

In this molecule, the sulfur atom is tetracoordinated, and the molecule contains 45 atoms, making option **C** correct.

CHECKPOINT

1 PTS

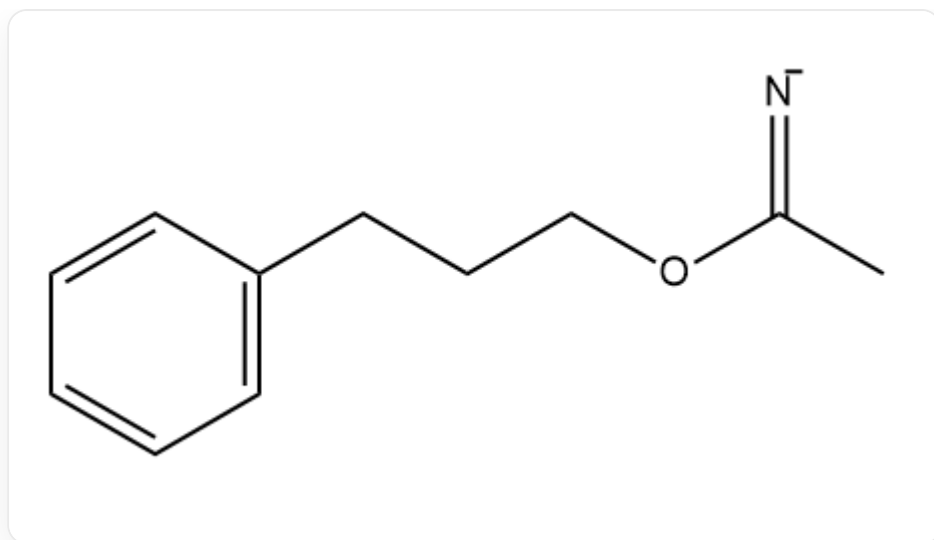
Contains 45 atoms

Finally, option **H** remains. If the solvent is acetonitrile, it may participate in the reaction. Acetonitrile is an electrophile, with its electrophilic site at the carbon of the cyano group, and can be attacked by the alkoxide to yield an intermediate:

CHECKPOINT

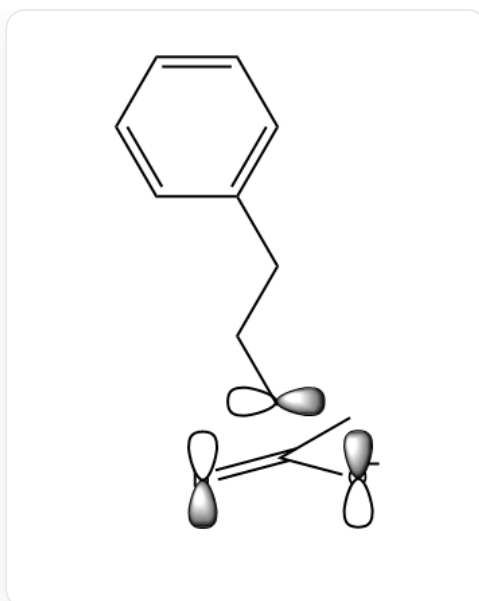
1 PTS

Cyanocarbon of acetonitrile attacked by alkoxide



CC(=[N-])OCCCC1CCCCC1

This intermediate then undergoes a $[1,3] - \sigma$ migration rearrangement. The carbon with partial positive charge migrates from the oxygen end to the nitrogen end via a suprafacial-antara $[1,3] - \sigma$ migration. The corresponding molecular orbitals are:



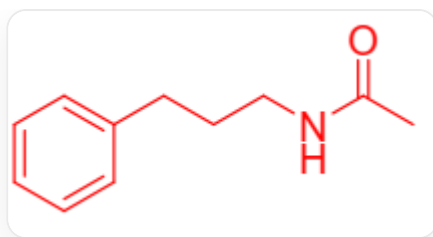
The figure shows three dumbbell shapes: two vertical dumbbells on the left and right sides, and one horizontal dumbbell above them. The left vertical dumbbell has a white upper lobe and a gradient gray lower lobe; the right vertical dumbbell has a gradient gray upper lobe and a white lower lobe. The horizontal dumbbell has a white left lobe and a gradient gray right lobe. From a central black point, two single lines and a pair of parallel double lines extend: the double lines point left-down to the middle of the left dumbbell; one single line points right-down to the middle of the right dumbbell; the other single line points right-up toward the gap between the right dumbbell and the upper horizontal dumbbell. A three-segment zigzag line extends upward from the horizontal dumbbell, connecting to a vertex of a hexagon. Inside the hexagon, lines parallel to its top, bottom-left, and bottom-right edges are present.

Subsequent protonation yields a stable amide product.

CHECKPOINT

1 PTS

[1, 3] – σ migration rearrangement occurs



CC(=O)NCCCCc1ccccc1

This product contains nitrogen, oxygen, and a benzene ring but lacks fluorine and sulfur, so option **H** is correct.

In summary, the only incorrect option is **G**.